

LibreOffice Calc Add-In: US Census Data (Java)

Claude Code prompt — UNO Add-In packaged as an .oxt extension

Prompt

Build a **LibreOffice Calc Add-In in Java** that exposes US Census Bureau data as worksheet functions, packaged as an **.oxt** extension. Target LibreOffice 24.8+ on Windows with the LibreOffice SDK installed.

Architecture (standard UNO Add-In pattern)

- Implement the add-in via `com.sun.star.sheet.AddIn`, with the service registered so functions are callable as `=CENSUS_...()` formulas.
- Provide the required interfaces: the custom add-in service interface, plus `XAddIn`, `XServiceName`, `XServiceInfo`, and `XLocalizable`.
- Use an `.idl`-defined interface (compiled with `idlc + regmerge`) OR the newer annotation approach — pick one and be consistent; document the build step either way.
- Return typed `Object[][]` arrays for spilled/range results, use proper Calc error values (do not let exceptions bubble to `#VALUE!` unexpectedly), and implement in-session caching keyed by request parameters to avoid hammering the API on recalc.
- **No third-party HTTP/JSON dependencies** — use `java.net.http.HttpClient` (JDK 11+) for requests and hand-roll or use only JDK-available JSON parsing. The Census API returns JSON as a 2D array of arrays, which maps cleanly to `Object[][]` with minimal parsing.

API details

- Census data API base: `https://api.census.gov/data`. Requests take the form `https://api.census.gov/data/{year}/{dataset}?get={variables}&for={geography}&key={API_KEY}`.
- The API key must be configurable — read it from an environment variable or a config file in the user profile, NOT hardcoded. Provide a `CENSUS_SETKEY`-style mechanism or document where the key file lives.
- Response format is a JSON array where the first row is headers and subsequent rows are data — map this directly to a spilled `Object[][]`.

Worksheet functions to expose

- `CENSUS_GET(year; dataset; variables; geography)` — core query, returns a spilled 2D range including the header row.
- `CENSUS_VALUE(year; dataset; variable; geography)` — single scalar value for one variable/geography.
- `CENSUS_VARLABEL(year; dataset; variable)` — human-readable label for a variable code (from the dataset's `variables.json`).
- `CENSUS_DATASETS(year)` — list available datasets for a year (from the discovery endpoint).

Include per-function descriptions and argument descriptions (via the `XAddIn` `display-name`/`description` methods) so they appear properly in the Calc Function Wizard.

Deliverables

- Full Java source, the service/interface definition, and a `MANIFEST/description.xml` for the `.oxt`.
- A build script (batch or shell) that compiles against the SDK, runs `idlc/regmerge` as needed, and packages the `.oxt`.
- A README covering: setting the API key, building on Windows, installing the extension, and 3–4 example formulas with real dataset/geography values (e.g. ACS 5-year population by state).

Note: LibreOffice Calc uses **semicolons** as argument separators — reflect that in all example formulas in the README and comments.